



Computational Genomics

DNA Sequencing and Data Analysis

Final Project - Group D

Long-read fecal microbiome analysis

Host-origin inference, microbiome profiling, and assembly of one dominant microbial species



The unknown sample resolves into four defensible findings

Cat

Best host match among six candidate animal genomes.

BMP6

Top annotated domestic-cat gene overlap, with an explicit MAPQ caveat.

34.35%

Host-unmapped reads assigned to *P. mirabilis* by Kraken2 clade count.

99.33%

Average dnadiff identity of the assembly against HI4320.

The strongest story is not one perfect signal; it is a chain of independent evidence: host alignment points to cat, host-unmapped taxonomy points to *Proteus mirabilis*, and assembly/reference comparison supports that microbial call.

A 15-minute presentation split exactly like the assignment

Methods - 5 min

Run reports, NanoComp/NanoPlot QC, filtering logic, tool choices, and reproducibility boundaries.

Bioinformatics - 5 min

Host alignment comparison, annotation overlap, Kraken2 taxonomy, read extraction, Flye assembly, and dnadiff.

Biological findings - 5 min

Domestic-cat origin, BMP6 as a host-region candidate, *Proteus mirabilis* dominance, and assembly caveats.

1

QC

What data do we trust?

2

Filter

Keep enough signal.

3

Host

Which animal?

4

Region

Which host locus?

5

Microbes

Who is in the sample?

6

Assembly

Can we recover a genome?

The sample is rich enough for discovery, but noisy enough to demand filtering

Metric	Group D FASTQ	Why it matters
Reads	1,803,479	Enough depth for screening and classification.
Total bases	1.98 Gb	Enough sequence for a pilot metagenomic assembly.
Median read length	512 bp	Many reads are short for ONT, so length filtering helps.
N50	2,297 bp	Longer reads still anchor alignments and assembly.
Mean quality	10.4	Supports Q10 as a practical floor.

Why MinKNOW plus NanoComp?

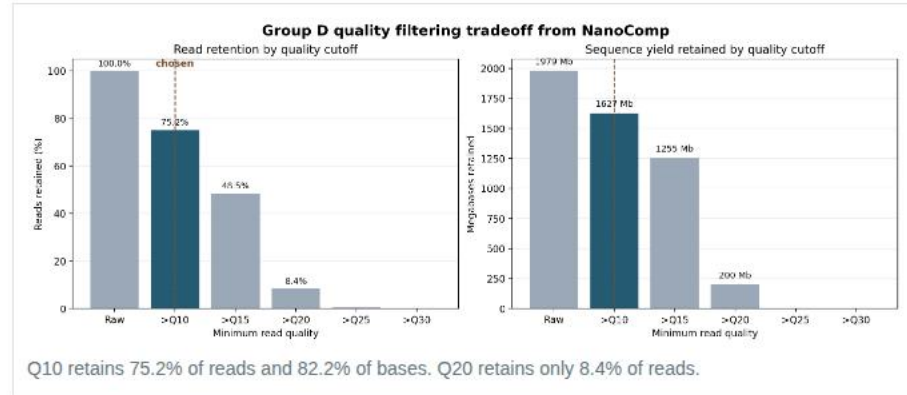
MinKNOW explains the sequencing run: device, flow cell, kit, basecalling, and run counters.

NanoComp/NanoPlot explain the analyzed FASTQ: read counts, bases, length, N50, and quality distribution.

The downstream workflow uses FASTQ-level evidence because those are the reads that enter alignment, classification, and assembly.

Filtering decision

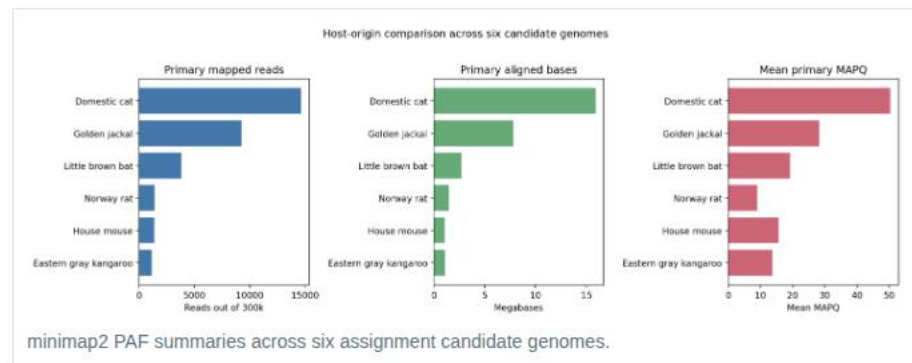
Q10 keeps the usable signal; Q15 and Q20 become too expensive



Cutoff	Reads	Bases	Decision
Raw	1.80 M	1.98 Gb	Baseline evidence.
>Q10	1.36 M	1.63 Gb	Chosen for main analysis.
>Q15	0.87 M	1.25 Gb	Cleaner but discards too much.
>Q20	0.15 M	0.20 Gb	Too sparse for broad exploration.

Final filter: $Q \geq 10$ and length ≥ 500 bp. The first 300,000 filtered reads were used as a reproducible pilot for the six-genome host comparison.

The host signal is low, but the relative winner is clear: domestic cat



Candidate	Primary mapped	Mean MAPQ
Domestic cat	14,669	50.34
Golden jackal	9,251	28.25
Little brown bat	3,818	19.20
Norway rat	1,393	9.05
House mouse	1,379	15.55
Eastern gray kangaroo	1,155	13.71

Fecal samples are mostly microbial, so host-mapping percentage is expected to be low. The decision comes from the consistent relative pattern across all candidates.

Why this host method?

Tool choices were made to match ONT reads and the assignment scale

minimap2 for host alignment

Chosen because map-ont is designed for noisy long reads, runs quickly on large genomes, and produces compact PAF summaries.

Why not BLAST or short-read aligners?

BLAST is too slow for six large genomes at this scale. Short-read aligners are less natural for noisy ONT read length/error profiles.

Decision metric

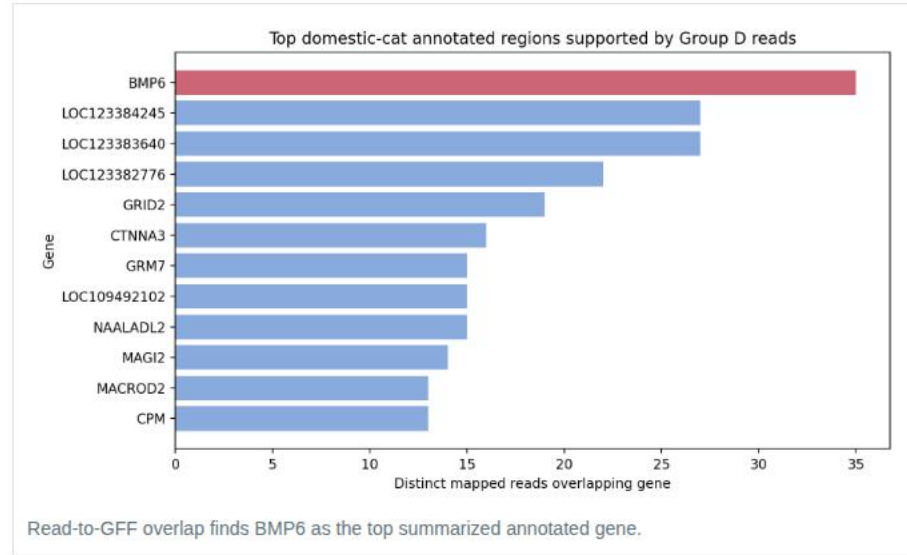
Use a combination of primary mapped reads, aligned bases, and mapping quality instead of a single count.

Interpretation guardrail

Low host fraction is not a failure. It is exactly what a fecal sample should look like when microbiome DNA dominates.

The assignment did not require a perfect host assembly; it required a reasoned comparative alignment decision. Domestic cat is the strongest supported host under that rule.

BMP6 is the best interpretable host-region candidate, but it needs a caveat



Biological role

BMP6 is a bone morphogenetic protein family member in TGF-beta signaling, with roles in development, skeletal biology, and iron/hepcidin regulation.

Why not a predicted LOC gene?

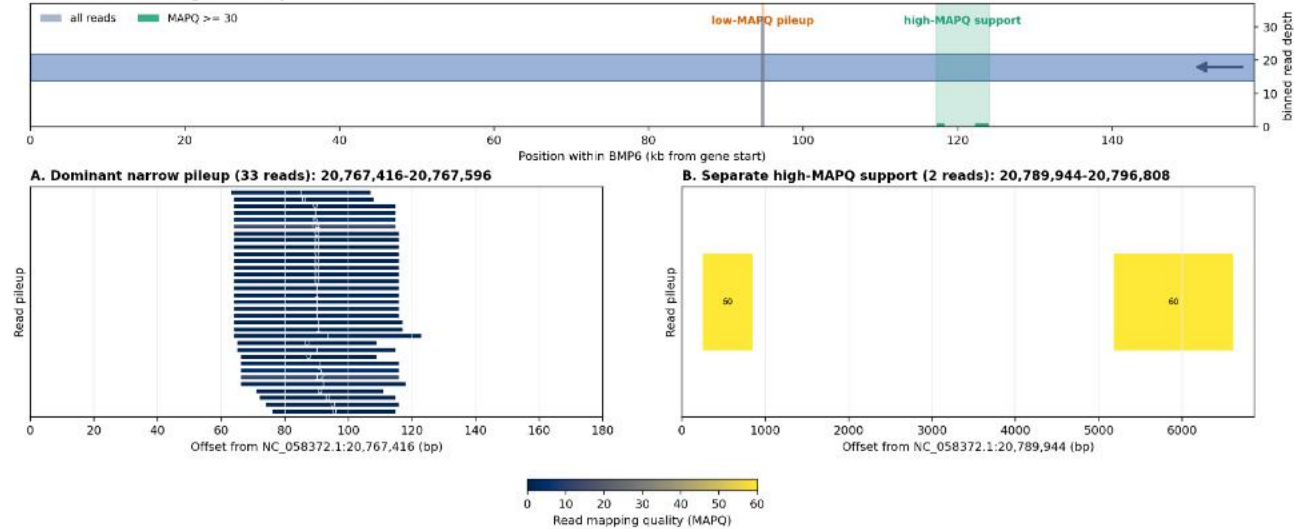
BMP6 has a recognizable biological function, making it a better presentation target than a high-count predicted locus with unclear interpretation.

The genome-browser view is where we stay honest

BMP6 host-region evidence: genome-browser-style view

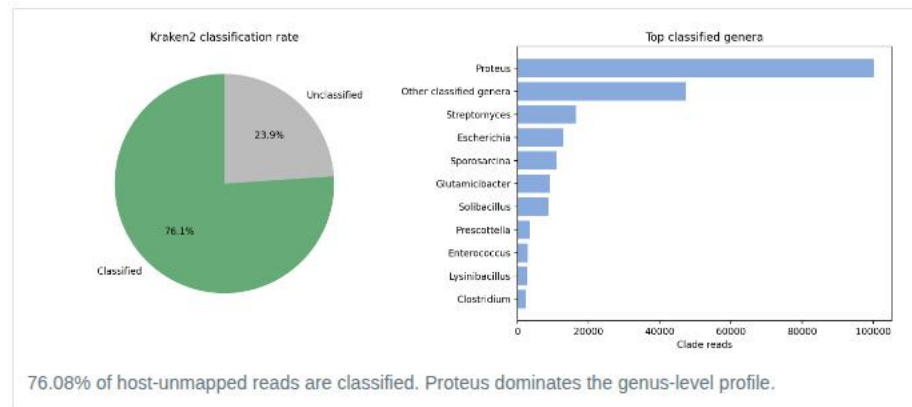
NC_058372.1:20,672,770-20,831,131 | 158.4 kb gene | 35 overlapping reads | 2 reads with MAPQ ≥ 30

Gene model, coverage, and inspected windows



Most BMP6-supporting reads form a narrow low-MAPQ pileup; two separate MAPQ-60 reads support a second interval. Good presentation claim: BMP6 is a candidate functional region, not complete high-confidence gene coverage.

Kraken2 classifies most host-unmapped reads and highlights Proteus



285k

Host-unmapped
reads profiled.

76.08%

Classified by
Kraken2.

35.11%

Proteus genus
clade reads.

23.92%

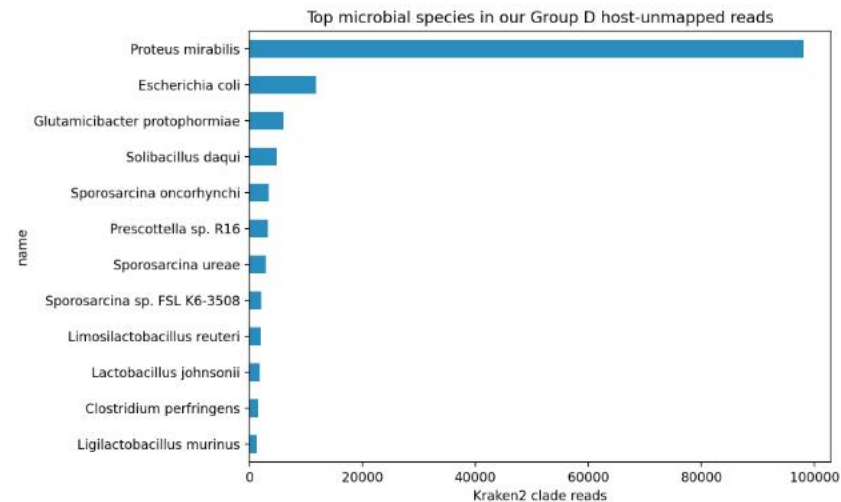
Unclassified reads.

Kraken2 was chosen over Centrifuge/MetaPhlAn/BLAST-style alternatives because it is fast, has compact public databases, and preserves read-level assignments for assembly extraction.

Proteus mirabilis is the clear assembly target

Species	Clade reads	Percent
<i>Proteus mirabilis</i>	98,025	34.35%
<i>Escherichia coli</i>	11,794	4.13%
<i>Glutamicibacter protophormiae</i>	6,029	2.11%
<i>Solibacillus daqui</i>	4,860	1.70%
<i>Sporosarcina oncorhynchi</i>	3,486	1.22%

It is abundant enough to extract reads, biologically plausible in a fecal context, and has public reference genomes for correctness assessment.



P. mirabilis is far above the next species-level calls.

Extract one abundant species, assemble it, then compare to a reference

1

Classify

Kraken2 on host-unmapped reads.

2

Choose

P. mirabilis as dominant species.

3

Extract

98,025 species/child-taxid reads.

4

Assemble

Flye for noisy ONT reads.

5

Reference

HI4320 from NCBI.

6

Validate

dnadiff identity and aligned coverage.

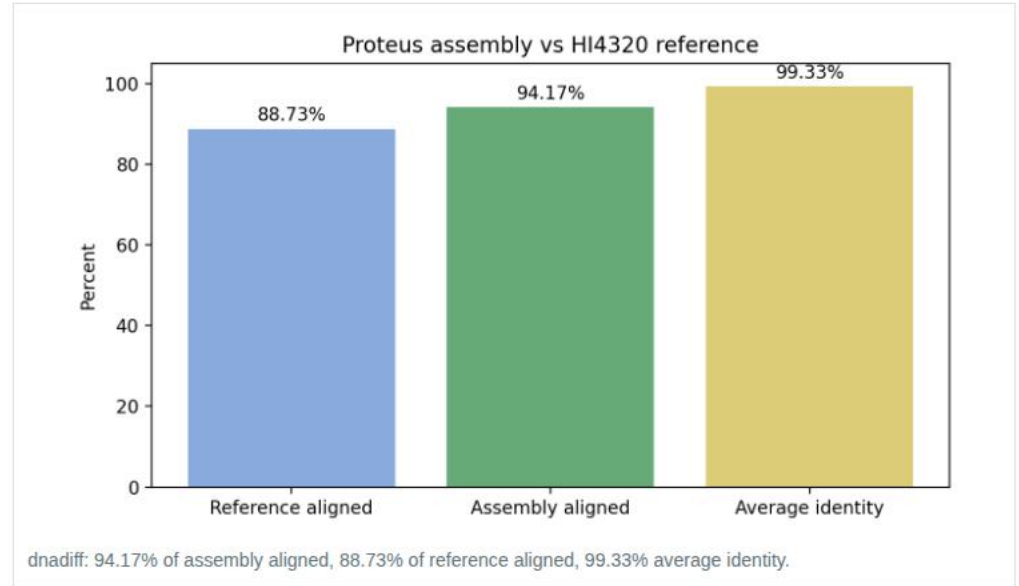
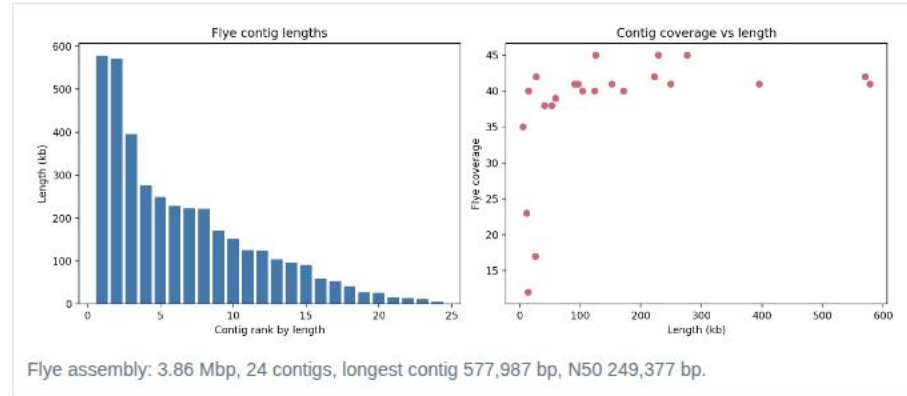
Why Flye?

Practical long-read assembler for noisy ONT data. Lighter for this notebook workflow than Canu, and more appropriate than short-read assemblers.

Why dnadiff?

Directly answers the assignment's correctness question with aligned coverage, identity, SNPs, and indels against a public reference.

The assembly is fragmented, but strongly reference-supported



This is not a closed isolate-grade genome. It is a metagenomic long-read species recovery with strong reference similarity.

What we learned from Group D's unknown sample

The most defensible biological story is: a domestic-cat fecal sample with a strong *Proteus mirabilis* signal, plus enough long-read evidence to recover a fragmented but high-identity assembly for the dominant microbe.

Technically

QC and filtering preserved enough signal for host comparison, taxonomy, and assembly.

Bioinformatically

Independent tools converged: minimap2 for host, GFF overlap for BMP6, Kraken2 for microbes, Flye plus dnadiff for assembly.

Biologically

The host and microbiome findings are plausible for fecal material, with caveats stated where evidence is limited.